A Deep Learning Approach to Stress Prediction Based on Sleep and Screen Time Patterns

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Abstract. Here will be the abstract… soon :))

I. INTRODUCTION

Same as the abstract :))

II. DATA DESCRIPTION

This dataset is called *Sleep, Screen Time and Stress Analysis*, obtained from Kaggle. It contains 15, 000 synthetic observations that aim to represent individuals with different lifestyles and habits.

Categorical features:

- **user_id:** Unique identifier per observation.
- **gender:** Male, Female or Other.
- **occupation:** Professional background of the individual.

Numerical features:

- **age:** Age of the individuals.
- **daily_screen_time_hours**: Total of hours using digital devices.
- **phone_usage_before_sleep_minutes** Minutes using digital devices before going to sleep.
- **sleep_duration_hours**: Average hours of sleep per night.
- **sleep_quality_score**: Score from 1 to 10 ranking sleep quality.
- **stress_level**: Measured stress in scale from 1 to 10
- **caffeine_intake_cups**: Caffeinated drinks during the day.
- **physical_activity_minutes**: Minutes spent on physical activity.
- **notifications_received_per_day**: Notifications from smartphone per day.
- **mental_fatigue_score**: Mental fatigue in scale from1 to 10.

III. PREPROCESSING

a. Transformation

No null observations were identified, but a column does not help the purpose of the model, as it does not provide meaningful information about the individual's behavior: the user id. Therefore, the column was removed.

The other two categorical features were encoded in order to become an input to the model. The *gender* feature had three values, as shown previously. Due to the low representation of the *Other* category (4%) in *gender*, it was excluded to prevent the model from bias during training. This decision is based on data limitations and is not related to gender diversity considerations. The *gender* feature turned into a binary variable, having 0 for Male and 1 for Female.

Occupation had 8 different values: Designer, Teacher, Software Engineer, Manager, Student, Freelancer, Doctor and Researcher. To prevent the model from interpreting them as ordinal values, one hot encoding was applied. Using *k - 1* dummy variables, multicollinearity between them was avoided. The value *Designer* was not included, meaning when all the dummy variables equal 0, it corresponds to *Designer*.
Except for the scores and the stress level, the numerical features were obtained in two scales: hours and minutes. To ensure a

consistent scale, all of them were converted to hours. Therefore, the column names were updated to reflect the correct units.

b. Data correlation

The correlation matrix shows that most of the features have weak or almost null linear relationships between themselves and the target value. This indicates that stress level is influenced by complex relationships with the features rather than just linear. Therefore, a deep learning model could help capture this pattern.

There are only three features with a relatively high linear relationship: mental fatigue score, quality sleep score and daily screen time in hours. Considering the nature of the situation, it makes sense that these have a stronger relationship with the stress.

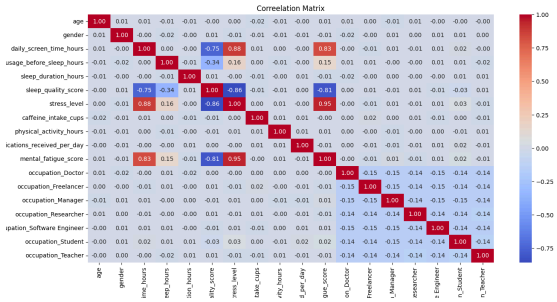

Figure 1. Correlation matrix

c. Data distribution

The target value stress level is skewed to the left with a higher concentration of observations near the maximum value. To improve training stability, both the target value and mental fatigue were normalized dividing by 10, leaving them in a range from 0 to 1. This change doesn't affect the imbalance. However, the impact is limited, facilitating learning and preventing bias from the model.

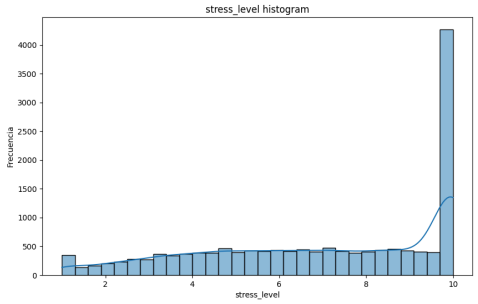

Figure 2. Stress level (target) histogram, skewed to the left.

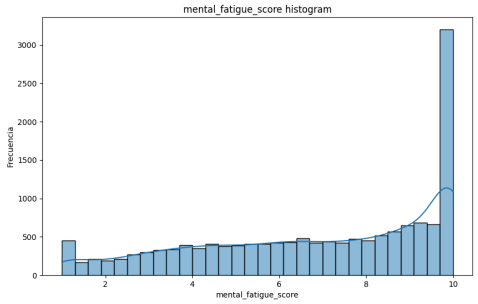

Figure 3. Mental fatigue score histogram, skewed to the left.

For the rest of the columns, they either have an almost perfect balance or have a normal distribution. There is no need to apply further processing.

d. Data split

The dataset does not have a previous separation for training and testing, Therefore, 70% of it will be dedicated to training, 15% to validation and 15% to test.

IV. MODEL

e. Model description

Description